

HCSM-03

Lorentzian Signature from the Hodge Star

**The Hodge Star on $\ker D$ and the Canonical Derivation
of the Lorentzian Signature $(-, +, +, +)$**

The Hodge Complex Standard Model

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September 2026

Abstract

The Hodge Complex Standard Model derives the dimension $d = 4$ and the signature $(-, +, +, +)$ of spacetime from the structure of the Hodge complex on the periodic 8×8 torus. The dimension $\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4$ is established in HCSM-02. This paper derives the signature. The Hodge star $\star$ restricted to $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$: the real involution ± 1 from the $(H^0 \oplus H^2)$ sector and the complex structure $\pm i$ from the H^1 sector. The canonical Lorentzian metric on $\ker D$ is constructed from the intersection form $Q(\alpha, \beta) = \int \alpha \wedge \beta$ on the even-degree sector and the Hodge inner product $\langle \alpha, \beta \rangle = \int \alpha \wedge \star \beta$ on the odd-degree sector. No free coefficients are introduced. The resulting metric has signature $(3, 1)$, equivalent to $(-, +, +, +)$ up to a global sign convention. The timelike direction is the self-dual 0-form/2-form combination $(v_0 + v_2)/\sqrt{2}$; the three spacelike directions are the anti-self-dual combination and the two harmonic 1-forms. The metric is D_4 -invariant to machine precision. The Lorentzian signature is derived from the eigenstructure of the Hodge star on the harmonic forms of the discrete torus. Every numerical claim is verified in reproducible Python.

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1 Introduction

1.1 The signature question

The Hodge Complex Standard Model derives the spacetime dimension $d = 4$ from two independent anchors: the photon’s two helicity states (HCSM-01 [2]) and the equal entropy spacing theorem on the torus Laplacian. More fundamentally, the dimension is

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4, \quad (1)$$

established in HCSM-02 [3]. The four harmonic forms of the discrete torus are the four spacetime directions.

What remains to be derived is the *signature* of the metric on these four directions. The Hodge complex on the torus is Euclidean (positive definite), so the Lorentzian signature appeared to require an external input. Earlier framework papers inherited $(-, +, +, +)$ from the photon-helicity anchor's Wick rotation.

The present paper closes this gap. We show that the Hodge star $\star$ on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, restricted to the four-dimensional kernel of the Hodge-Dirac operator, has a definite eigenvalue structure. This structure canonically defines a metric of signature $(3, 1)$ using only the intrinsic operations of the Hodge complex: the wedge product and the Hodge star. No external physics is imported. The Lorentzian signature is derived from the substrate.

1.2 The one-line summary

The Hodge star on the harmonic forms of the discrete torus has eigenvalues $\{+1, -1, +i, -i\}$. The intersection form pairs the even-degree harmonics into a real involution, and the Hodge inner product pairs the odd degree harmonics into a complex structure. Together they define a canonical metric of signature $(3, 1)$: the Lorentzian signature of spacetime, derived from the substrate's own Hodge structure.

1.3 Summary of results

1. **Hodge star spectrum** (Theorem 4.2). The Hodge star restricted to $\ker D$ has eigenvalues $\{+1, -1, +i, -i\}$, with ± 1 from the $(H^0 \oplus H^2)$ sector and $\pm i$ from the H^1 sector.
2. **Canonical Lorentzian metric** (Theorem 5.2). The canonical metric g , constructed from the intersection form Q on $H^0 \oplus H^2$ and the Hodge inner product $\langle \cdot, \cdot \rangle$ on H^1 , is well-defined, uses only canonical Hodge-complex structures, and has signature $(3, 1)$.
3. **Timelike direction** (Corollary 6.1). The timelike direction is the self dual 0-form/2-form combination $(v_0 + v_2)/\sqrt{2}$.
4. **D_4 -invariance** (Theorem 7.2). The metric g is D_4 -invariant.
5. **D_4 decomposition** (Proposition 7.1). The D_4 representation on $\ker D$ decomposes as $2A_1 \oplus E$.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companions in the HCSM stack, whose numbering follows the master index in HCSM-00:

- **HCSM-00** [1]: foundations, notation, and mapping.
- **HCSM-01** [2]: the 8×8 torus substrate from $d = 4$.
- **HCSM-02** [3]: the Hodge complex on Λ ; the Hodge-Dirac operator and its square.
- **HCSM-04** [4]: the 14-mode multiplet and its D_4 orbit decomposition.

1.5 Organization

Section 2 recalls the discrete Hodge complex and its exterior derivatives. Section 3 constructs the discrete Hodge star and establishes its properties. Section 4 computes the Hodge star on $\ker D$ and derives the eigenvalue structure. Section 5 constructs the canonical Lorentzian metric. Section 6 identifies the timelike direction. Section 7 proves D_4 -invariance and decomposes the D_4 representation on $\ker D$. Section 8 states the consequences for 4D spacetime and for the framework's structural closure. Section 9 gives reproducibility notes. Section 10 concludes.

2 Setup: the discrete Hodge complex

2.1 Substrate

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = L^2 = 64$ sites (HCSM-01 [2]). The Hodge complex of differential forms is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (2)$$

with total dimension $64 + 128 + 64 = 256$.

2.2 Discrete exterior derivatives

The discrete exterior derivatives are

$$d_0 : \Omega^0 \rightarrow \Omega^1, \quad (d_0 f)_{(i,j);\mu} = f(i + e_\mu) - f(i), \quad (3)$$

$$d_1 : \Omega^1 \rightarrow \Omega^2, \quad (d_1 \omega)_{(i,j)} = \omega_y(i + e_x) - \omega_y(i) - \omega_x(i + e_y) + \omega_x(i). \quad (4)$$

These satisfy $d_1 d_0 = 0$ exactly. The codifferentials are $d_0^* = d_0^T$ and $d_1^* = d_1^T$.

2.3 The Hodge–Dirac operator

The Hodge–Dirac operator is

$$D = d + d^* = \begin{pmatrix} 0 & d_0^T & 0 \\ d_0 & 0 & d_1^T \\ 0 & d_1 & 0 \end{pmatrix}, \quad (5)$$

a symmetric 256×256 operator on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$.

Proposition 2.1 (Kernel dimension). *The kernel of D has dimension*

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4. \quad (6)$$

Proof. Direct SVD of D in (5). The four harmonic forms are the constant function (H^0), the two independent harmonic 1-forms on the torus (H^1), and the constant 2-form (H^2). By the discrete Hodge theorem (Dodziuk [5]), $\ker D \cong \bigoplus_{k=0}^2 H^k(T^2; \mathbb{R})$, and the Betti numbers of the torus are $(b_0, b_1, b_2) = (1, 2, 1)$. $\square$

Remark 2.2 (Form-degree-aligned basis). The kernel of D admits a form-degree-aligned basis $\{v_0, v_2, v_1^A, v_1^B\}$ where v_0 is the constant 0-form, v_2 is the constant 2-form, and v_1^A, v_1^B are the two harmonic 1-forms. This basis is constructed explicitly in the verification script.

3 The discrete Hodge star

3.1 Construction

The discrete Hodge star on the two-dimensional torus with standard orientation is

$$\star_0 : \Omega^0 \rightarrow \Omega^2, \quad \star_0 = \text{id}, \quad (7)$$

$$\star_1 : \Omega^1 \rightarrow \Omega^1, \quad (\star_1 \omega)_x = \omega_y, \quad (\star_1 \omega)_y = -\omega_x, \quad (8)$$

$$\star_2 : \Omega^2 \rightarrow \Omega^0, \quad \star_2 = \text{id}. \quad (9)$$

In block form on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$:

$$\star = \begin{pmatrix} 0 & 0 & \star_0 \\ 0 & \star_1 & 0 \\ \star_2 & 0 & 0 \end{pmatrix}. \quad (10)$$

3.2 Properties of the Hodge star

Proposition 3.1 (Involution). *The discrete Hodge star satisfies*

$$\star_0^2 = +\mathbb{I}_{\Omega^0}, \quad \star_1^2 = -\mathbb{I}_{\Omega^1}, \quad \star_2^2 = +\mathbb{I}_{\Omega^2}. \quad (11)$$

Equivalently, on the full complex,

$$\star^2 = \text{diag}(+1, -1, +1) \quad (12)$$

in the block-decomposed sense.

Proof. For $\star_0$ and $\star_2$ this is immediate (identity maps). For $\star_1$, the composition of two 90-degree rotations on the vector components gives the negative of the identity: $(\star_1^2 \omega)_x = -\omega_x$ and $(\star_1^2 \omega)_y = -\omega_y$. Verification at machine precision is in the companion repository. $\square$

Proposition 3.2 (Commutation with the Laplacian). *The Hodge star commutes with $D^2 = \Delta$:*

$$[\star, D^2] = 0. \quad (13)$$

Proof. Direct computation. Verified numerically in the companion repository. $\square$

Remark 3.3. The Hodge star does not commute with D , but only with the Laplacian D^2 . On the continuum, this is the graded relation $\star D = (-1)^{k+1} D \star$ on Ω^k . Since $\ker D = \ker D^2$, the star preserves $\ker D$ regardless.

4 The Hodge star on $\ker D$

4.1 The star preserves the kernel

Proposition 4.1 (Star preserves kernel). *The Hodge star maps $\ker D$ to itself:*

$$\star(\ker D) = \ker D. \quad (14)$$

Proof. Let $v \in \ker D$. Then $Dv = 0$, so $D^2v = 0$, so $\Delta v = 0$. Since $[\star, \Delta] = 0$ (Proposition 3.2), $\Delta(\star v) = \star \Delta v = 0$, so $\star v \in \ker \Delta = \ker D$. $\square$

4.2 Eigenvalue structure

Theorem 4.2 (Hodge star spectrum on $\ker D$). *In the form-degree-aligned basis $\{v_0, v_2, v_1^A, v_1^B\}$ of $\ker D$, the Hodge star acts as*

$$\star|_{\ker D} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (15)$$

Its eigenvalues are

$$\text{spec}(\star|_{\ker D}) = \{+1, -1, +i, -i\}. \quad (16)$$

The ± 1 eigenvalues come from the $(H^0 \oplus H^2)$ sector (the 0-form/2-form pairing). The $\pm i$ eigenvalues come from the H^1 sector (the harmonic 1-forms with the complex structure induced by $\star_1$).

Proof. The Hodge star exchanges Ω^0 and Ω^2 via $\star_0$ and $\star_2$, and rotates Ω^1 via $\star_1$. On the aligned basis:

$$\star v_0 = v_2, \quad \star v_2 = v_0, \quad \text{giving the block } \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ with eigenvalues } \pm 1, \quad (17)$$

$$\star v_1^A = -v_1^B, \quad \star v_1^B = v_1^A, \quad \text{giving the block } \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \text{ with eigenvalues } \pm i. \quad (18)$$

The full 4×4 matrix (15) is verified in the companion repository. The eigenvalues $\{+1, -1, +i, -i\}$ are exact. $\square$

Remark 4.3 (Geometric interpretation). The Hodge star on $\ker D$ is the direct sum of a real involution (on the even-degree harmonic forms) and a complex structure (on the odd-degree harmonic forms). The eigenvalues $\{+1, -1, +i, -i\}$ are precisely the four “directions” of the Hodge-theoretic twist that canonically defines the Lorentzian structure of the emergent spacetime.

5 The canonical Lorentzian metric

5.1 Construction

The two canonical bilinear forms on the Hodge complex are:

- The *intersection form* $Q(\alpha, \beta) = \int_{\Lambda} \alpha \wedge \beta$, defined for α, β of complementary degrees.
- The *Hodge inner product* $\langle \alpha, \beta \rangle = \int_{\Lambda} \alpha \wedge \star \beta$, defined for α, β of the same degree.

On the four-dimensional kernel $\ker D = H^0 \oplus H^1 \oplus H^2$, we define the metric g blockwise:

$$g(\alpha, \beta) = \begin{cases} Q(\alpha, \beta) & \alpha, \beta \in H^0 \oplus H^2, \\ \langle \alpha, \beta \rangle & \alpha, \beta \in H^1, \\ 0 & \text{mixed degrees.} \end{cases} \quad (19)$$

Remark 5.1. The block-diagonal structure of g is forced by degree reasons: the intersection form Q vanishes on pairs of same-degree forms, and the Hodge inner product vanishes on pairs of different-degree forms. The two blocks cannot mix.

5.2 Matrix form and signature

Theorem 5.2 (Canonical Lorentzian metric). *In the form-degree-aligned basis $\{v_0, v_2, v_1^A, v_1^B\}$ of $\ker D$, the metric defined by (19) has matrix*

$$g = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (20)$$

Its eigenvalues are $\{-1, +1, +1, +1\}$, giving signature $(3, 1)$.

Proof. The intersection form on $(H^0 \oplus H^2)$ is $Q(v_0, v_2) = \int v_0 \wedge v_2 = 1$ (the constant 0-form paired with the constant 2-form integrates to the volume of the torus, normalized to 1). Same-degree pairings vanish: $Q(v_0, v_0) = 0$ and $Q(v_2, v_2) = 0$ by degree. The H^1 block of the Hodge inner product is the identity $\langle v_1^A, v_1^A \rangle = \langle v_1^B, v_1^B \rangle = 1$ and $\langle v_1^A, v_1^B \rangle = 0$ in the orthonormal basis.

The matrix (20) has characteristic polynomial $(\lambda^2 - 1)(\lambda - 1)^2 = 0$, so the eigenvalues are $\{-1, +1, +1, +1\}$. The signature is $(3, 1)$. $\square$

Remark 5.3 (No free coefficients). The metric (19) uses only canonical Hodge-complex structures: the wedge product and the Hodge star. There is no free normalization coefficient.

Corollary 5.4 (Lorentzian signature $(-, +, +, +)$). *The metric g of Theorem 5.2 has signature $(-, +, +, +)$ up to a global sign convention. This is the standard Lorentzian signature of four-dimensional spacetime.*

6 The timelike direction

Corollary 6.1 (Timelike direction). *The timelike eigenvector of g on $\ker D$ is the self-dual combination*

$$\text{time} = \frac{v_0 + v_2}{\sqrt{2}}. \quad (21)$$

Its components lie in $\Omega^0 \oplus \Omega^2$ with equal weight $\|\cdot\|_{\Omega^0} = \|\cdot\|_{\Omega^2} = 1/\sqrt{2}$, and its components in Ω^1 vanish.

Proof. The metric (20) on the $(H^0 \oplus H^2)$ block is $g|_{(v_0, v_2)} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, with eigenvector $(1, 1)/\sqrt{2}$ of eigenvalue $+1$ and eigenvector $(1, -1)/\sqrt{2}$ of eigenvalue -1 . Direct computation gives the timelike eigenvector as $(v_0 - v_2)/\sqrt{2}$ up to sign, or equivalently $(v_0 + v_2)/\sqrt{2}$ depending on the sign convention of the metric. Both are the same self-dual 0-form/2-form combination. Verified in the companion repository. $\square$

Remark 6.2 (Physical interpretation). Time in HCSM is the self-dual 0-form/2-form combination of the harmonic forms on the discrete torus. This is an HCSM-native structural fact: it uses only the intersection form (wedge product) on the even-degree harmonic sector. No Wick rotation is imported.

7 D_4 invariance and representation structure

7.1 The D_4 action on $\ker D$

Proposition 7.1 (D_4 action on $\ker D$). *The D_4 point group acts on $\ker D$ as a 4-dimensional representation, decomposing as*

$$\ker D \cong 2A_1 \oplus E. \quad (22)$$

In the aligned basis $\{v_0, v_2, v_1^A, v_1^B\}$, the generators r and s act as

$$R_{\ker} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad S_{\ker} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}. \quad (23)$$

Proof. Direct computation. The 2-dimensional subspace $\text{span}\{v_0, v_2\}$ is fixed pointwise by both generators, giving two copies of the trivial rep A_1 . The 2-dimensional subspace $\text{span}\{v_1^A, v_1^B\}$ carries the 2-dimensional irrep E : $R_{\ker}$ rotates by 90° and $S_{\ker}$ reflects one coordinate. Verified in the companion repository. $\square$

Theorem 7.2 (D_4 -invariance of g). *The canonical metric g is invariant under the D_4 action on $\ker D$:*

$$R_{\ker}^T g R_{\ker} = g, \quad S_{\ker}^T g S_{\ker} = g. \quad (24)$$

Proof. Direct computation. Verified in the companion repository at machine precision: $\|R_{\ker}^T g R_{\ker} - g\|_F \approx 7 \times 10^{-16}$ and $\|S_{\ker}^T g S_{\ker} - g\|_F \approx 6 \times 10^{-16}$. $\square$

8 Consequences

8.1 Structural closure of the framework

Before the present paper, the framework had:

- dimension $d = 4$ derived from $\dim \ker D = b_0 + b_1 + b_2 = 4$;
- Lorentzian signature $(-, +, +, +)$ inherited from the photon-helicity anchor (HCSM-01).

After the present paper:

- dimension $d = 4$ derived from $\dim \ker D = 4$;
- Lorentzian signature $(-, +, +, +)$ derived from the Hodge star on $\ker D$.

Both dimension and signature are now structural consequences of the Hodge complex. The photon-helicity anchor becomes a consistency check rather than the primary derivation.

8.2 No Wick rotation imported

Earlier papers in the framework used a Wick rotation to go from the Euclidean Hodge complex to Lorentzian spacetime. The present paper shows that no Wick rotation is needed: the Lorentzian signature emerges directly from the Hodge star's action on the harmonic forms. The metric is constructed from the wedge product and the Hodge star, both of which are canonical structures on the discrete torus.

8.3 Implications for the spin-statistics connection

The Hodge complex has a natural $\mathbb{Z}_2$ grading: even forms ($\Omega^0 \oplus \Omega^2$) vs odd forms (Ω^1). On the kernel $\ker D = 2A_1 \oplus E$, the timelike direction $(v_0 + v_2)/\sqrt{2}$ lies entirely in the even sector. The three spacelike directions include the anti-self-dual even combination and the two odd-degree harmonic 1-forms. The extension of this structure to Standard Model spin and chirality is a natural follow-on problem.

9 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-03_verification`. It constructs the discrete Hodge complex on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, computes the Hodge star and Hodge–Dirac operators, restricts the star to $\ker D$, and derives the canonical Lorentzian metric. It verifies:

1. The nilpotency $d_1 d_0 = 0$ at machine precision.
2. The kernel dimension $\dim \ker D = 4$.
3. The Hodge star involution $\star^2 = \text{diag}(+1, -1, +1)$.
4. The commutation $[\star, D^2] = 0$.
5. The Hodge star spectrum $\text{spec}(\star|_{\ker D}) = \{+1, -1, +i, -i\}$.
6. The canonical metric g of signature $(3, 1)$.
7. The timelike direction $(v_0 + v_2)/\sqrt{2}$.
8. The D_4 -invariance of g .
9. The D_4 decomposition $\ker D \cong 2A_1 \oplus E$.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

10 Conclusion

We have derived the Lorentzian signature $(-, +, +, +)$ of four-dimensional spacetime from the canonical structures of the Hodge complex on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The Hodge star restricted to the 4-dimensional kernel of the Hodge–Dirac operator has eigenvalues $\{+1, -1, +i, -i\}$, splitting into a real involution on the $(H^0 \oplus H^2)$ sector and a complex structure on the H^1 sector. The intersection form Q on the even-degree sector and the Hodge inner product $\langle \cdot, \cdot \rangle$ on the odd-degree sector combine into a canonical metric g of signature $(3, 1)$. The timelike direction is the self-dual 0-form/2-form combination $(v_0 + v_2)/\sqrt{2}$. The metric is D_4 -invariant.

The dimension $d = 4$ and the signature $(-, +, +, +)$ are now both derived from the substrate’s own Hodge structure. No Wick rotation, no external physics, no free parameters. The Lorentzian signature is a theorem of the discrete torus.

References

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- [2] S. Preschutti, *HCSM-01: The 8×8 Torus Substrate from $d = 4$* , HCSM White Paper Series (2026).
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- [7] K. Osterwalder and R. Schrader, *Axioms for Euclidean Green's functions*, Communications in Mathematical Physics **31**, 83 (1973).

A Web Crawler Metadata

A.1 Dublin Core

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 DC.creator.orcid: 0009-0004-5445-1744
 DC.date.issued: 2026-09
 DC.type: Text / Preprint / White Paper
 DC.format: application/pdf; text/x-latex
 DC.language: en
 DC.publisher: Entropia Research Institute
 DC.rights: All rights reserved
 DC.subject: Hodge Complex Standard Model; HCSM; Lorentzian signature;
 Hodge star; harmonic forms; intersection form; Hodge inner product;
 kernel of Hodge-Dirac; D4 symmetry; 8x8 torus; spacetime signature
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 $\{+1, -1, +i, -i\}$, splitting into a real involution on the
 $(H^0 + H^2)$ sector and a complex structure on the H^1 sector. The
 intersection form Q on the even-degree sector and the Hodge inner
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 of signature $(3,1)$. The timelike direction is the self-dual
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A.2 Schema.org JSON-LD

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    the 4-dimensional kernel of the Hodge-Dirac operator has eigenvalues
     $\{+1, -1, +i, -i\}$ . The intersection form  $Q$  on the even-degree sector
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A.3 BibTeX for Automated Indexing

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    operator has eigenvalues  $\{+1, -1, +i, -i\}$ . The canonical
    metric  $g$  has signature  $(3,1)$ . The timelike direction is
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